Akshay Tirunelveli Sriram

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Education

National Institute of Technology, Calicut — Calicut, India
Bachelor of Technology in Electronics and Communication Engineering — *Expected Graduation: 2027*

- Coursework: Machine Learning, Predictive and Generative AI, Computer Vision/Deep Learning, NLP, Reinforcement Learning, Statistics and Probability, DSA, OOP

Technical Skills

Languages/Libraries:	Python, SQL, Flask, FastAPI
ML/Computer Vision:	PyTorch, OpenCV, NumPy, Pandas, Matplotlib, Power BI
Cloud/Platforms:	AWS, Git, Jupyter Notebook, n8n
Specializations:	Computer Vision, Image Restoration, Anomaly Detection, Agentic AI Workflows, Real-Time Inference, Automation, Data Preprocessing, Feature Engineering, Statistical Analysis

Experience

Machine Learning Engineer — Remote
Kednus Corporation · Part-time — *Aug 2026 – Present*

- Build automated, multi-step agentic workflows that collect evidence and map controls across the audit lifecycle, replacing manual review passes with logged, reproducible agent runs.
- Develop Python evaluation harnesses that detect bias, drift, and unsafe outputs in deployed models, converting raw model behavior into structured audit artifacts aligned to EU AI Act and ISO 42001 control requirements.

Research Intern — Bloomington, IL
Illinois Wesleyan University · Prof. Mark Liffiton — *June 2026 – Aug 2026*

- Researched LLM guardrailing for educational AI (RLHF, Constitutional AI, prompt engineering, PII-preserving data handling), synthesizing findings into a five-layer guardrail framework that informed CodeHelp's system prompt architecture.
- Shipped merged pull requests to Gen-Ed (Flask/SQLite/Python): an instructor-authored tutor opening message system that eliminated redundant per-student LLM API calls (cutting cost and latency), and a class activity summarization pipeline with anonymized student IDs for FERPA compliance.

Projects

AquaVision – Deep Learning Image Restoration for Degraded Visual Conditions | *Python, PyTorch, OpenCV, NumPy* GitHub

- Built an attention-residual U-Net that restores images degraded by scattering and wavelength-dependent attenuation, the same physics behind atmospheric haze and low-visibility camera feeds; outperformed U-Net and FUnIE-GAN baselines by 2.5 dB PSNR on the UIEB benchmark at 9.1 ms/frame inference, suitable for real-time pipelines.

MindBridge – Agentic AI Mental Health Assistant | *Python, FastAPI, LangGraph, ChromaDB, React* GitHub

- Built a multi-agent LangGraph system (emotion, cognitive, empathetic agents routed by a meta-agent) with ChromaDB RAG memory and crisis detection scoring, using Hugging Face datasets amod/mental_health_counseling_conversations and vibhorag101/phr_mental_therapy_dataset; deployed via FastAPI and React.

TalentIQ – AI Resume Shortlisting System | *Python, Streamlit, Mistral AI* GitHub

- Built an end-to-end AI pipeline that parses resumes, scores candidates on semantic similarity, achievements, and ownership, and deploys a Streamlit UI with explainable tier classification.

Leadership

Team Lead – AI/ML Division — Feb 2025 – Present
AquaRobotics Club, NIT Calicut

- Co-founded the club and led the inauguration demo of a lifesaving robot; now preparing the team for national AUV competitions.

Student Point of Contact – IEEE YESS 2025 Marketing Team — Jul 2025 – Sept 2025

- Led marketing strategy contributing to 10 lakhs in revenue, collaborating across technical and business teams.

Technical Coordinator – COMSOC Society, IEEE NITC — Feb 2025 – Present

- Ran hands-on ML/data science workshops for student members, owning curriculum and session delivery.

Vice Captain – College Tennis Team — Feb 2024 – Present